

Shrinivas Ramasubramanian

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EDUCATION

- **Carnegie Mellon University** Pittsburgh, PA
MS in Robotics; GPA: 4.17/4.0 Aug. 2026 (*expected*)
- **Indian Institute of Technology Bombay** Mumbai, India
B.Tech in Electrical Engineering; GPA: 7.42/10.0 Aug. 2020

PUBLICATIONS

- **Tail-Likelihood Reinforcement Learning.**
Shrinivas Ramasubramanian, Daman Arora, Fahim Tajwar, Guanning Zeng, Qingyang Wu, Zhongzhu Zhou, Chenfeng Xu, Haiwen Feng, Yuda Song, Aarti Singh, Ruslan Salakhutdinov, J. Andrew Bagnell, Jeff Schneider, Andrea Zanette.
arXiv preprint, 2026. [PDF]
- **Improving Model-Based Reinforcement Learning by Converging to Flatter Minima.**
Shrinivas Ramasubramanian, Ben Freed, Alexandre Capone, Jeff Schneider.
NeurIPS 2025 (Main Conference). [PDF]
- **Selective Mixup Fine-Tuning for Optimizing Non-Decomposable Metrics.**
Shrinivas Ramasubramanian[†], Harsh Rangwani[†], Sho Takemori[†], Kunal Samanta, Kato Takashi, Yuhei Umeda, Venkatesh Babu Radhakrishnan.
ICLR 2024, Spotlight (Main Conference). [PDF]
- **Cost Sensitive Self-Training For Optimizing Non-Decomposable Objectives.**
Harsh Rangwani[†], Shrinivas Ramasubramanian[†], Sho Takemori[†], Kato Takashi, Yuhei Umeda, Venkatesh Babu Radhakrishnan.
NeurIPS 2022 (Main Conference). [PDF]
- **Cost-sensitive learning for long-tailed temporal action segmentation.**
Zhanzhong Pang, Fadime Sener, Shrinivas Ramasubramanian, Angela Yao.
BMVC 2024 (Main Conference). [PDF]
- **Long-Tail Temporal Action Segmentation via Group-wise Temporal Logit Adjustment.**
Zhanzhong Pang, Fadime Sener, Shrinivas Ramasubramanian, Angela Yao.
ECCV 2024 (Main Conference). [PDF]

[†] Equal contribution

PATENTS

- **Information processing apparatus and machine learning method.**
Harsh Rangwani, Shrinivas Ramasubramanian, Sho Takemori, Kato Takashi, Yuhei Umeda, Venkatesh Babu Radhakrishnan.
US Patent: US20230376846A1.
- **Machine Learning method and information processing apparatus.**
Shrinivas Ramasubramanian, Harsh Rangwani, Kunal Samantha, Sho Takemori, Yuhei Umeda, Venkatesh Babu Radhakrishnan.
US Patent: US20250037023A1.

ACADEMIC SERVICE

- **Teaching Assistant DS 265o, IISc:** Primary TA for the course Deep-Learning for Computer Vision DS-265o at IISc Fall'22 , a master's-level course.
- **Reviewer:** NeurIPS 2023, ICML 2023, ICLR 2024, AAAI 2024, NeurIPS 2025.

WORK EXPERIENCE

- **Research Engineer** Bangalore, India
Fujitsu Research India Pvt. Ltd. May 2023 – Present
 - **Robust person re-identification:** Worked on fine-tuning CLIP for robust person re-identification under distribution shifts.
 - **Temporal graph learning for fraud detection:** Developed consistent sub-sampling methods for large (84M+ nodes) heterogeneous temporal graphs of financial data to train and evaluate PoCs for fraud detection.
 - **Non-decomposable objectives:** Continued work on optimizing non-decomposable and non-differentiable objectives.
- **Research Intern (Prof. Angela Yao)** Remote / Bangalore, India
CVML Group, National University of Singapore Mar. 2022 – May 2023
 - **Long-tailed action segmentation:** Applied expertise in long-tail learning to enhance action segmentation in videos, collaborating with Prof. Angela Yao's group.
 - **Loss design and architectures:** Adapted established architectures and designed novel loss functions, significantly improving metrics for imbalanced action segmentation; work published at **ECCV 2024** and **BMVC 2024**.
- **Project Assistant (Prof. Venkatesh Babu)** Bangalore, India
Video Analytics Lab (VAL), IISc Bangalore Oct. 2021 – May 2023
 - **NeurIPS 2022 publication:** Worked on optimizing **linear non-decomposable objectives** such as bias and fairness measures for long-tailed data using **semi-supervised learning**, resulting in a **NeurIPS 2022** publication in collaboration with **Fujitsu Research**.
 - **Teaching Assistant:** Prepared academic material and delivered lectures on **Semi-supervised Learning, Object Detection, and Generative Models** for DS-265.
 - **Mentoring and ICLR Spotlight:** Mentored summer interns and undergraduates and co-authored “**Selective Mixup Fine-Tuning for Optimizing Non-Decomposable Metrics**”, published as an **ICLR 2024 Spotlight**, addressing both non-linear and linear confusion-matrix objectives under semi-supervised and supervised settings.
- **Computer Vision Engineer** Bangalore, India
Beltech Jan. 2021 – Oct. 2021
 - **Automatic License Plate Recognition:** Built and deployed an ALPR pipeline: fine-tuned CRAFT for multi-line plate text, added a Spatial Transformer + LSTM OCR, and template-based correction.
 - **Intersection occupancy and violations:** Fused multi-camera detections with static satellite maps via affine projection to generate intersection occupancy grids and implemented heuristic traffic-violation detection.
 - **Multi-object tracking:** Deployed a YOLOv4 + DeepSort multi-object tracking system and designed track-level heuristics, reaching mAP of 0.71 on held-out intersections.

ACHIEVEMENTS

- **NeurIPS 2022 paper invited talk:** Paper “Cost Sensitive Self-Training For Optimizing Non-Decomposable Objectives” presented at the IISc-EECS Research Students Symposium.
- **JEE Advanced:** All-India Rank 156 in **JEE Advanced** among ~150k aspirants; received a scholarship of INR 95k from Vidyamandir Classes (VMC).

REFERENCES

Dr. Venkatesh Babu

Professor, CDS Department, IISc Bangalore
Website: <http://cds.iisc.ac.in/faculty/venky/>
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Dr. Angela Yao

Assistant Professor, School of Computing, National University of Singapore
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Email: ayao@comp.nus.edu.sg

Dr. Yuhei Umeda

Senior Project Director, Fujitsu AI Lab, Japan
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Dr. Jeff Schneider

Professor, Carnegie Mellon University
Website: <https://www.cs.cmu.edu/~schneide/>
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